Network flows exercises

A09-F01

Audit the nonnegative-edge-flow model. Original uv has capacity 7 and flow 5; vu has capacity 4 and flow 1. List both residual channels directed u→v, with capacities, and explain why aggregation needs an update rule.

A09-F02

After round 2 of the six-vertex example, give every residual on P_3, its bottleneck, and all resulting original-edge flows. Verify conservation at x and y.

A09-F03

Prove $f(S,T)=|f|\leq c(S,T)$ for every s–t cut. Explain why cut capacity does not subtract T→S capacities.

A09-F04

Recover the residual reachable cut after termination. Give its capacity, the flow value, and the optimality argument. Also compute the capacity for S={s,w}.

A09-F05

Prove equivalence of no augmenting path and existence of a cut whose capacity equals the flow value. Explicitly handle original T→S edges.

A09-F06

In the four-vertex U example take U=8. How many alternating bad-path augmentations occur? Why is O(EF*) not polynomial in binary input length? Distinguish rational from arbitrary real capacities.

A09-F07

Prove distance monotonicity for Edmonds–Karp. If a new shortest path uses newly appearing u→v, why does the old labeling satisfy d(u)=d(v)+1?

A09-F08

Derive the full $O(VE^2)$ Edmonds–Karp bound beyond distance monotonicity. Explain critical-arc recurrence, event count, per-round cost, and capacity assumptions.

A09-P01 | Diagnose the initial state

After initialization e(s)=-10 and e(t)=0. Is this a feasible flow of value 10? Give the active set.

A09-P02 | Check push conditions

At step 0, a→c has residual capacity 3 and e(a)=6. Why is push illegal? What does the next legal relabel produce?

A09-P03 | Compute a reverse push

Starting at step 11, calculate δ for a→s, the updated f(s,a), e(s), and e(a). Explain why h(s) stays fixed.

A09-P04 | Reconstruct current arcs

Reconstruct every pointer advance and relabel reset in the first discharge of a, steps 1 through 14. Does a push immediately advance current?

A09-P05 | Reconstruct the list

Continue from step 25 with L=[b,a,c]. Give discharge order and moves through termination. Which CLRS invariant or implementation rule changes if only active vertices are kept?

A09-P06 | Complete the correctness proof

Prove that valid heights exclude a residual s-to-t path. Explain why intermediate excesses must still be zero before applying the maximum-flow theorem.

A09-P07 | Counts and running time

Derive bounds for relabels, saturating pushes, and nonsaturating pushes. State the potential and preprocessing needed for n=O(m). Why is a relabel not O(1)?

A09-P08 | Certify the flow and scheduling bound

Compute every final original-edge flow and an equal-capacity cut. Explain how the $O(n^3)$ relabel-to-front proof uses at most one nonsaturating push per discharge.

A09-M01

Build the network for E={a1,a2,b1,c2,c3}. Give three unit paths, a size-3 matching, and an optimality cut.

A09-M02

Prove integral-flow extraction. Use a fractional maximum flow in $K_{2,2}$ to refute the claim that every maximum flow is integral.

A09-M03

For E={a1,b1,c2,c3}, give a maximum matching, a capacity-2 cut, and a subset witnessing failure to match all of L.

A09-M04

Split v in s→v=5, v→t=5 with vertex capacity 3. Find maximum flow, prove conservation after merging, and give size when k vertices are split.

A09-M05

In the multisource example, require supplies 3,2 and sink quotas 3,2. Prove infeasibility using supernodes. What is the unrestricted total optimum?

A09-M06

For A→B:[2,4], B→C:[1,3], C→A:[0,3], derive each d, build the auxiliary network, and recover a circulation.

A09-M07

Change C→A to [0,1]. Give a deficient auxiliary cut and explain the contradiction in the original graph.

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A09-M08

Test exact values 2 and 3 for s→a:[1,3], a→t:[1,2]. Explain a safe finite return capacity and when an integral solution is guaranteed.

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd ed., Chapter 26, Maximum Flow. Original handwritten algorithms notes (演算法_wjungle), network-flow pages. MIT 6.046J, Fall 2004, Lectures 20 and 21, Maximum Flow and Edmonds–Karp.

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